

Tarea dección 4.4

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1297

$$3) \lim_{x \rightarrow a} f(x) = 0$$

$$\lim_{x \rightarrow a} g(x) = 0$$

$$\lim_{x \rightarrow a} h(x) = 7$$

$$\lim_{x \rightarrow a} p(x) = \infty$$

$$\lim_{x \rightarrow a} q(x) = \infty$$

$$a) \lim_{x \rightarrow a} [f(x) - p(x)] = \lim_{x \rightarrow a} f(x) - \lim_{x \rightarrow a} p(x) = 0 - \infty = -\infty$$

$$b) \lim_{x \rightarrow a} [p(x) - q(x)] = \lim_{x \rightarrow a} p(x) - \lim_{x \rightarrow a} q(x) = \infty - \infty = \text{for. indeterminada}$$

$$c) \lim_{x \rightarrow a} [p(x) + q(x)] = \lim_{x \rightarrow a} p(x) + \lim_{x \rightarrow a} q(x) = \infty + \infty = \infty$$

$$7) \lim_{x \rightarrow 1} \frac{x^3 - 2x^2 + 1}{x^3 - 1} = \frac{1^3 - 2(1)^2 + 1}{1^3 - 1} = \frac{0}{0} \text{ for. indeterminada}$$

$$\lim_{x \rightarrow 1} \frac{3x^2 - 4x}{3x^2} = \frac{3(1)^2 - 4(1)}{3(1)^2} = -\frac{1}{3}$$

$$13) \lim_{x \rightarrow (\pi/2)^-} \frac{\cos x}{1 - \sin x} = \frac{\cos \frac{\pi}{2}}{1 - \sin \frac{\pi}{2}} = \frac{0}{0} \text{ for. indeterminada}$$

$$\lim_{x \rightarrow (\pi/2)^+} \frac{\sin x}{\cos x} = -\infty$$

$$15) \lim_{t \rightarrow 0} \frac{e^{2t} - 1}{\sin t} = \frac{e^{2(0)} - 1}{\sin 0} = \frac{0}{0} \text{ for. indeterminada}$$

$$\lim_{t \rightarrow 0} 2e^{2t} = 2$$

$$19) \lim_{x \rightarrow \infty} \frac{\ln x}{\sqrt{x}} = \frac{\infty}{\infty}$$

$$\lim_{x \rightarrow \infty} \frac{\frac{1}{x}}{\frac{1}{2\sqrt{x}}} = \frac{2\sqrt{x}}{x} = \frac{2\sqrt{x}}{\infty} = \frac{\infty}{\infty}$$

$$\lim_{x \rightarrow \infty} \frac{2 + \frac{1}{2\sqrt{x}} + x - 2\sqrt{x} + 1}{x^2} = \frac{1}{\sqrt{x}x} = \frac{1}{\infty} = 0$$

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$$23) \lim_{t \rightarrow 1} \frac{t^8 - 1}{t^5 - 1} = \frac{1-1}{1-1} = 0$$

$$\lim_{t \rightarrow 1} \frac{8t^7}{5t^4} = \frac{8(1)^7}{5(1)^4} = \frac{8}{5}$$

$$25) \lim_{x \rightarrow 0} \frac{\sqrt{7+2x} - \sqrt{7-4x}}{x} = \frac{\sqrt{7+2(0)} - \sqrt{7-4(0)}}{0} = \frac{1-1}{0} = \frac{0}{0}$$

$$\lim_{x \rightarrow 0} \frac{1}{2\sqrt{7+2x}} \cdot 2 - \frac{1}{2\sqrt{7-4x}} \cdot (-2) = \frac{1}{\sqrt{7+2x}} + \frac{2}{\sqrt{7-4x}} = \frac{1}{1} + \frac{2}{1} = 3$$

$$27) \lim_{x \rightarrow 0} \frac{e^x - 1 - x}{x^2} = \frac{e^0 - 1 - 0}{0^2} = \frac{1-1-0}{0} = \frac{0}{0}$$

$$\lim_{x \rightarrow 0} \frac{e^x - 1}{2x} = \frac{e^0 - 1}{2(0)} = \frac{1-1}{0} = \frac{0}{0}$$

$$\lim_{x \rightarrow 0} \frac{e^x}{2} = \frac{e^0}{2} = \frac{1}{2}$$

$$33) \lim_{x \rightarrow 0} \frac{x 3^x}{3^x - 1} = \frac{0 \cdot 3^0}{3^0 - 1} = \frac{0}{1-1} = \frac{0}{0}$$

$$\lim_{x \rightarrow 0} \frac{1(3^x) + x(\ln(3))(3^x)}{(\ln(3)) 3^x} = \frac{3^0(\ln(3)) 3^0}{(\ln(3)) 3^0} = 1$$

$$41) \lim_{x \rightarrow 1} \frac{x^a - ax + a - 1}{(x-1)^2} = \frac{ax^{a-1} - a}{2x-2} = \frac{(a-1)a}{2}$$

$$\lim_{x \rightarrow 1} \frac{(a-1)a}{2} = \frac{1}{2} a(a-1)$$

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$$47) \lim_{x \rightarrow \infty} x^3 e^{-x^2} = \infty^3 e^{-\infty} = \infty \cdot 0 = \infty$$

$$\lim_{x \rightarrow \infty} x^3 e^{x^2} + x^3 e^{-x^2}$$

$$\lim_{x \rightarrow \infty} 3x^2 e^{x^2} + x^3 e^{-x^2} - 2x$$

$$\lim_{x \rightarrow \infty} 3x^2 e^{x^2} - 2x^4 e^{-x^2} = \lim_{x \rightarrow \infty} 3x^2 e^{x^2} - 2x^4 e^{-x^2} = 0$$

$$51) \lim_{x \rightarrow 1} \left(\frac{x}{x-1} - \frac{1}{\ln x} \right) = \frac{1}{x-1} - \frac{1}{\ln 1} = \infty - \infty$$

$$\lim_{x \rightarrow 1} \frac{(\ln x)(x) - (x-1)}{(\ln x)(x-1)} = \frac{(\ln(x))x - x + 1}{(\ln(x))(x-1)} = \frac{(\ln(1))1 - 1 + 1}{(\ln(1))(1-1)} = \frac{0}{0}$$

$$\lim_{x \rightarrow 1} \frac{\frac{1}{x} * x + (\ln(x)) - 1 + 0}{\frac{1}{x}(x-1) + (\ln(x))1} = \frac{-(\ln(x))}{1 - \frac{1}{x} + (\ln(x))} = \frac{0}{0}$$

$$\lim_{x \rightarrow 1} \frac{\frac{1}{x}}{\frac{1+x}{x^2}} = \frac{x}{1+x} = \frac{1}{1+1} = \frac{1}{2}$$

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$$59) \lim_{x \rightarrow 0} (1-2x)^{\frac{1}{x}} = (1-2 \cdot 0)^{\frac{1}{0}} = 1^{\frac{1}{0}} = 1^{\infty}$$

$$\ln y = \ln(1-2x)^{\frac{1}{x}}$$

$$\ln y = \frac{1}{x} \left(\frac{\ln(1-2x)}{1} \right)$$

$$\lim_{x \rightarrow 0} \ln y = \lim_{x \rightarrow 0} (1) \left(\frac{\ln(1-2x)}{x} \right)$$

$$\ln \lim_{x \rightarrow 0} y = \lim_{x \rightarrow 0} \frac{\ln(1-2x)}{x} = \frac{0}{0}$$

$$\ln \lim_{x \rightarrow 0} y = \lim_{x \rightarrow 0} \frac{-2}{1} = -2$$

$$\lim_{x \rightarrow 0} y = e^{-2}$$

$$63) \lim_{x \rightarrow \infty} x^{\frac{1}{x}} = \infty^{\frac{1}{\infty}} = \infty^0$$

$$\ln y = \ln x^{\frac{1}{x}}$$

$$\ln y = \left(\frac{1}{x} \right) \frac{\ln x}{1}$$

$$\lim_{x \rightarrow \infty} \ln y = \lim_{x \rightarrow \infty} 1 \left(\frac{\ln x}{x} \right)$$

$$\ln \lim_{x \rightarrow \infty} y = (1) \lim_{x \rightarrow \infty} \frac{\ln x}{x} = \frac{\ln \infty}{\infty} = \frac{\infty}{\infty}$$

$$\ln \lim_{x \rightarrow \infty} y = (1) \lim_{x \rightarrow \infty} \frac{\frac{1}{x}}{1} = \frac{1}{x}$$

$$\ln \lim_{x \rightarrow \infty} y = \frac{1}{x}$$

$$\lim_{x \rightarrow \infty} y = e^{\frac{1}{x}} = e^{\frac{1}{\infty}} = e^0 = 1$$

$$(65) \lim_{x \rightarrow 0^+} (4x+1)^{\cot x} = (4 \cdot 0 + 1)^{\cot(0)} = 1^\infty$$

$$\ln y = \ln (4x+1)^{\cot x}$$

$$\ln y = \cot(x) \left(\frac{\ln(4x+1)}{1} \right)$$

$$\lim_{x \rightarrow 0^+} \ln y = \lim_{x \rightarrow 0^+} \cot(x) \ln(4x+1)$$

$$\ln \lim_{x \rightarrow 0} y = \cot x \lim_{x \rightarrow 0} \ln(4x+1) = 0 \cdot \infty = 0$$

$$\ln \lim_{x \rightarrow 0} y = \cot x \lim_{x \rightarrow 0} \frac{1}{4x+1} \cdot 4 = \frac{4}{4x+1}$$

$$\ln \lim_{x \rightarrow 0} y = \cot x \left(\frac{4}{4x+1} \right)$$

$$\lim_{x \rightarrow 0^+} y = e^{\frac{4}{4x+1}} = e^{\frac{4}{1}} = e^4$$

$$(67) \lim_{x \rightarrow 0^+} (1 + \sin 3x)^{\frac{1}{x}} = (1 + \sin 3 \cdot 0)^{\frac{1}{0}} = 1^\infty$$

$$\ln y = \ln (1 + \sin 3x)^{\frac{1}{x}}$$

$$\ln y = \frac{1}{x} \cdot \frac{\ln 1 + \sin 3x}{1}$$

$$\lim_{x \rightarrow 0} \ln y = \lim_{x \rightarrow 0} (1) \left(\frac{\ln 1 + \sin 3x}{x} \right)$$

$$\ln \lim_{x \rightarrow 0} y = (1) \lim_{x \rightarrow 0} \frac{\ln 1 + \sin 3x}{x} = \frac{\ln 1 + \sin 3 \cdot 0}{0} = \frac{0}{0}$$

$$\ln \lim_{x \rightarrow 0} y = (1) \lim_{x \rightarrow 0} \frac{0 + \cos(3x) \cdot 3}{1} = 3 \cos 3x$$

$$\ln \lim_{x \rightarrow 0} y = (1) (3 \cos 3x)$$

$$\lim_{x \rightarrow 0} y = e^{3 \cos 3x} = e^{3 \cos 3 \cdot 0} = e^3$$